Scope Delta Analysis: The Meridian Pivot

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Project: Solstice Events Co. Check-in Kiosk (Sprint 2)

1. The Mid-Sprint Pivot

- **Original Requirement:** Build an event check-in kiosk that synchronously calls a badge-printer's REST API and waits for a success response before displaying "Checked In".
- **New Requirement:** Deprecate the synchronous API due to vendor changes. Rebuild using an asynchronous model by publishing requests to a message queue and exposing a webhook endpoint for completion callbacks.
- **Deadline Impact:** Zero extension granted; the 48-hour delivery mandate was maintained.

2. The Scope Delta

- **DROPPED - Synchronous API Logic:** The original synchronous REST API call code was marked deprecated and visibly removed to prevent obsolete code from running in parallel.
- **MODIFIED - Kiosk UI State:** Modified the UI so that it no longer shows "Checked In" instantly. It now enforces a "Pending" state upon scan until the webhook confirmation arrives.
- **ADDED - Message Queue:** Implemented an Azure Service Bus to handle incoming print requests asynchronously using a new technology.
- **ADDED - Webhook & Security:** Provisioned an Azure Function webhook to receive asynchronous vendor callbacks. Configured CORS for cross-domain security and maintained strict duplicate-scan protection under the new out-of-order model.

3. Regression Check

- **Test Executed:** Verified that the duplicate-scan protection still holds under the new asynchronous model by passing three distinct attendees (including one duplicate) through the webhook.
- **Status:** Pass. The architectural integrity was maintained; the UI correctly transitions from Pending to Checked In, and duplicates are successfully blocked.

4. Reprioritized Backlog

- **Advanced UI Animations:** Deprioritized custom CSS modal pop-ups to allocate sprint capacity to Azure webhook routing.
- **Offline Fallback:** Abandoned local browser-storage fallback to focus entirely on the mandatory webhook integration and state validation.